Unit 3 Review: Absolute Value Inequalities

Ms. Okafor’s Algebra 1 · Test on Friday · fictional practice packet

Absolute value

Absolute value: the distance of a number from zero on the number line, so it is never negative. For example $|-4| = 4$ and $|4| = 4$.

Solving $|x| < a$

Absolute value less than a number: when $|x| < a$ with $a > 0$, the solutions are all numbers closer to zero than a, so $-a < x < a$. This is an “and” statement: both inequalities must be true at once.

Example: $|x - 3| < 5$ means $-5 < x - 3 < 5$. Add 3 to every part to get $-2 < x < 8$.

Solving $|x| > a$

Absolute value greater than a number: when $|x| > a$ with $a > 0$, the solutions are farther from zero than a, so $x < -a$ or $x > a$. This is an “or” statement.

Example: $|2x + 1| \geq 7$ means $2x + 1 \leq -7$ or $2x + 1 \geq 7$, so $x \leq -4$ or $x \geq 3$.

Page 2: Notation and special cases

Interval notation

Interval notation: a way to write a set of solutions. Parentheses mean the endpoint is not included (< or >); square brackets mean it is included (≤ or ≥). The solution $-2 < x < 8$ is written $(-2, 8)$. The solution $x \le -4$ or $x \ge 3$ is written $(-\infty, -4] \cup [3, \infty)$.

Special cases

- $|x| < -2$ has no solution, because an absolute value can never be less than a negative number.
- $|x| > -2$ is true for every real number, because an absolute value is always at least 0.

Common mistakes

- Writing $|x| > 3$ as $-3 > x > 3$, which is impossible. It should be $x < -3$ or $x > 3$.
- Forgetting to flip the inequality sign when multiplying or dividing by a negative number.
- Not isolating the absolute value first: in $2|x| + 1 < 9$, subtract 1 and divide by 2 before splitting.

Area connection: the area of a square with side x is x^2, so $x^2 < 9$ has the same solutions as $|x| < 3$.

Page 3: Practice problems (no answers given)

1. Solve $|x + 2| < 6$ and write the answer in interval notation.
2. Solve $|3x - 1| \geq 8$.
3. Solve $2|x| - 3 < 7$.
4. Explain why $|x| < -1$ has no solutions.